NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

SEMESTER: I

DEPARTMENT

OF

COMPUTER SCIENCE AND ENGINEERING

PRACTICE WORK BOOK

SUBJECT CODE: BE01R00111

SUBJECT NAME: BASIC ELECTRONICS ENGINEERING

Subject Code:	BE01R00111	**Subject Name**	**Credit:**	4
Semester:	I	Basic Electronics Engineering	**Hours:**	120
Department:		Computer Science and Engineering		

CHAPTER: 1
SEMICONDUCTOR DIODES

p-n junction diode, Characteristics and parameters, Diode approximations, DC load line analysis, Temperature effects, Diode AC models, Diode specifications, Diode testing, Zener diodes.

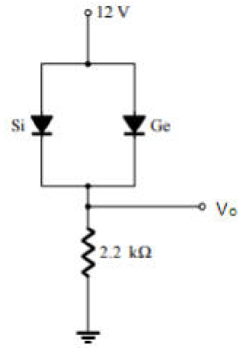
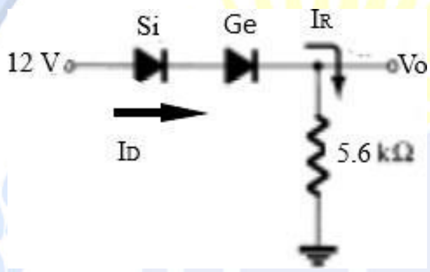
SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Differentiate between insulator, conductor and semiconductor. **(June-2019) [NLJIET]**	3	CO1	AN
2	Define between insulator, conductor and semiconductor. **(Mar-2023) [NLJIET]**	3	CO1	RM
3	What is a diode? Write its types and applications. **(Aug-2022) [NLJIET]**	3	CO1	RM
4	Define the p-n junction diode and draw its characteristics. **(June-2025-New) [NLJIET]**	3	CO1	RM
5	Discuss about VI characteristic of Ideal Diode. **(Jan-2020) [NLJIET]**	3	CO1	AN
6	Define Bulk Resistance **[NLJIET]**	3	CO1	RM
7	Explain the effect of temperature on the diode characteristics **[NLJIET]**	3	CO1	UN
8	Explain Zener diode. **(Apr-2022) [NLJIET]**	3	CO1	UN
9	State the difference between Avalanche breakdown and Zener breakdown. **(Jan-2024) [NLJIET]**	3	CO1	RM
10	Explain working of Zener break down. **(Jan-2025-New) [NLJIET]**	3	CO1	UN
11	Explain forward bias PN junction diode with diagram. **(June-2019) (Jan-2025-New) [NLJIET]**	4	CO1	UN
12	Discuss forward and reverse bias operation of a P-N junction diode with depletion region. **(Jan-2024) [NLJIET]**	4	CO1	AN
13	Draw and explain V-I characteristic of P-N junction diode. **(July-2024) [NLJIET]**	4	CO1	CR

14	Explain the diode V-I characteristics of ideal and practical PN junction semiconductor diode. **(Aug-2022) [NLJIET]**	**4**	**CO1**	**UN**
15	Explain DC load line Analysis of Diode **[NLJIET]**	**4**	**CO1**	**UN**
16	Draw and explain AC model of a diode. **[NLJIET]**	**4**	**CO1**	**CR**
17	State different types of diodes. Describe process of testing diode with multi meter. **(June-2019) [NLJIET]**	**4**	**CO1**	**RM**
18	Discuss different types of diodes. Describe process of testing diode with multi meter. **(Mar-2023) [NLJIET]**	**4**	**CO1**	**AN**
19	Draw the symbol and explain the working of zener diode. **[NLJIET]**	**4**	**CO1**	**CR**
20	What is zener breakdown? What is avalanche breakdown? Compare both the type of breakdown. **(Jan-2019) (Mar-2023) [NLJIET]**	**4**	**CO1**	**RM**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Write a short note: V-I characteristic of P-N junction diode. **(Jan-2019) (Mar-2023) (Jan-2025-New) [NLJIET]**	**7**	**CO1**	**UN**
2	Describe diode approximations in detail. **(Jan-2024) [NLJIET]**	**7**	**CO1**	**RM**
3	Draw and explain V-I characteristics of an ideal diode. **(Apr-2022) [NLJIET]**	**7**	**CO1**	**CR**
4	Derive the expressions for load line analysis in a diode circuit. **(June-2025-New) [NLJIET]**	**7**	**CO1**	**EL**
5	Comparison between P-N junction Diode and Zener Diode. **(July-2024) [NLJIET]**	**7**	**CO1**	**AN**
6	What is break down diode? Explain working of zener break down and avalanche break down **(June-2019) [NLJIET]**	**7**	**CO1**	**RM**

7	Determine the voltage Vo for the network of Figure 2. Give explanation for your answer. **(Jan-2020) [NLJIET]**  Figure 2	**7**	**CO1**	**EL**
8	Determine Vo and ID for the series circuit **[NLJIET]** 	**7**	**CO1**	**AP**

CHAPTER: 2
DIODE APPLICATIONS

Half-wave and Full-wave rectifiers, Power supply, RC and LC power supply filters, Zener diode voltage regulators, Series and shunt clipping circuits, Clamping circuits, DC voltage multipliers.

SHORT QUESTIONS

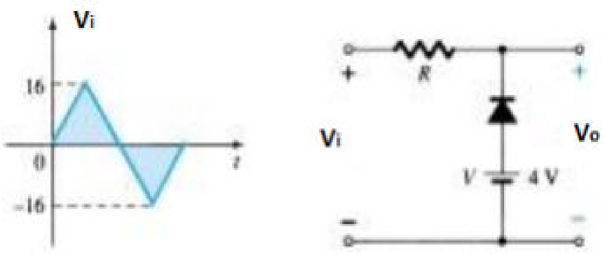
Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Draw the circuit diagram of Half wave rectifier. **(Jan-2020) [NLJIET]**	**3**	**CO1**	**CR**
2	Draw circuit diagram, waveform of a half-wave rectifier. **(June-2025-New) [NLJIET]**	**3**	**CO1**	**CR**
3	Why Zener diode can be used as voltage regulator? **(Jan-2025-New) [NLJIET]**	**3**	**CO1**	**UN**

4	Explain zener diode as a Voltage regulator. **(Aug-2023) [NLJIET]**	**3**	**CO1**	**UN**
5	Explain clipping circuit **(June-2019) [NLJIET]**	**3**	**CO1**	**UN**
6	Explain series positive clipper with diagram. **(July-2024) [NLJIET]**	**3**	**CO1**	**UN**
7	Explain Negative Clamping circuit with diagram. **(July-2024) [NLJIET]**	**3**	**CO1**	**UN**
8	Describe Negative clamper circuit with necessary waveforms **(Jan-2024) [NLJIET]**	**3**	**CO1**	**AN**
9	Draw voltage multiplier circuit. **(Mar-2023) (Jan-2020) (Jan-2025-New) [NLJIET]**	**3**	**CO1**	**CR**
10	Draw circuit for Voltage doubler and explain its operation. **(Jan-2024) [NLJIET]**	**3**	**CO1**	**CR**
11	Compare Half wave rectifier and Full wave bridge rectifier for following parameters. (i) Average DC Voltage (ii) Ripple Factor (iii) PIV (iv) Rectifier Efficiency **(Aug-2023) [NLJIET]**	**4**	**CO1**	**UN**
12	Explain the bridge rectifier with diagrams. **(Jan-2020) [NLJIET]**	**4**	**CO1**	**UN**
13	Explain advantage and disadvantage of bridge rectifier over full wave rectifier. **(Jan-2025-New) [NLJIET]**	**4**	**CO1**	**UN**
14	Describe RC filter with its limitations. **(Jan-2024) [NLJIET]**	**4**	**CO1**	**RM**
15	What is the purpose of filter? Explain RC Filter and LC Filter with waveforms. **[NLJIET]**	**4**	**CO1**	**UN**
16	Why Zener diode can be used as voltage regulator? Explain Zener as voltage regulator with necessary diagram. **(Jan-2020) [NLJIET]**	**4**	**CO1**	**UN**
17	Discuss Zener diode as voltage regulator. **(Jan-2024) [NLJIET]**	**4**	**CO1**	**AN**
18	Explain the working of a Zener diode as a voltage regulator. **(June-2025-New) [NLJIET]**	**4**	**CO1**	**UN**

19	Explain one series and one shunt clipping circuits with necessary circuit diagram and waveform. **(June-2025-New) [NLJIET]**	4	CO1	UN
20	Explain voltage multiplier circuits. **(Apr-2022) [NLJIET]**	4	CO1	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain half wave rectifier with neat diagram and also draw the output waveform. **(Jan-2025-New) [NLJIET]**	7	CO1	UN
2	Explain working full-wave rectifier with necessary waveforms. **[NLJIET]**	7	CO1	UN
3	Explain the working of a full-wave rectifier with a center-tapped transformer. **(June-2025-New) [NLJIET]**	7	CO1	UN
4	Draw circuit diagram of Center-taped full wave rectifier and explain its operation with necessary waveforms and derivation of Vdc. **(Jan-2024) [NLJIET]**	7	CO1	CR
5	Explain the working of a full-wave bridge rectifier with necessary circuit diagram and waveform. **(June-2025-New) [NLJIET]**	7	CO1	UN
6	Draw and Explain bridge rectifier. Explain advantage and disadvantage of bridge rectifier over full wave rectifier **(Jan-2019) (July-2024) [NLJIET]**	7	CO1	CR
7	Explain full wave bridge rectifier with neat diagram. **(Mar-2023) (June-2019) [NLJIET]**	7	CO1	UN
8	Explain full wave Bridge rectifier with neat circuit diagram and show the waveforms. **(Apr-2022) [NLJIET]**	7	CO1	UN
9	Give the advantages of Full wave bridge rectifier over centre tap rectifier. Explain bridge rectifier with neat diagram. Also draw load voltage, load current, diode voltage waveforms. **[NLJIET]**	7	CO1	CR
10	State the applications of Rectifier, and Comparison of Halfwave, Full-wave center- tap and Full-wave Bridge rectifier. **(July-2024) [NLJIET]**	7	CO1	RM

11	Explain the Zener Diode and how it works as the zener voltage regulator. **[NLJIET]**	**7**	**CO1**	**UN**
12	Draw and explain series and shunt positive clipper with output waveforms. **[NLJIET]**	**7**	**CO1**	**CR**
13	Explain the shunt and series clipping circuits with diagram. **(Jan-2025-New) [NLJIET]**	**7**	**CO1**	**UN**
14	Enumerate the different types of clipping circuits with their different names and input-output waveforms. **(Aug-2022) [NLJIET]**	**7**	**CO1**	**EL**
15	Explain the following diode clipping circuits with input and output waveforms. i) Series Negative Clipper, (ii) Parallel Positive Clipper **[NLJIET]**	**7**	**CO1**	**UN**
16	Illustrate the need of clipper circuit and explain Positive clipper and negative clipper with necessary diagram and waveforms. **(Aug-2023) [NLJIET]**	**7**	**CO1**	**EL**
17	Explain parallel clipper circuit in detail. **[NLJIET]**	**7**	**CO1**	**UN**
18	Write a short note: Biased clipper circuit. **(Jan-2019) [NLJIET]**	**7**	**CO1**	**AN**
19	Explain in details clippers and clampers. **(Apr-2022) [NLJIET]**	**7**	**CO1**	**UN**
20	Write short note on clamper. **[NLJIET]**	**7**	**CO1**	**AN**
21	Explain the difference between clipping and clamping circuit. A positive voltage clamping circuit and a positive voltage clipping circuit each have ±12 V square Wave input. Sketch the output waveform for each circuit. **(Jan-2020) [NLJIET]**	**7**	**CO1**	**UN**
22	Determine the Vo for the network shown in figure 1 **(Jan-2020) [NLJIET]**  Figure 1	**7**	**CO1**	**EL**

23	Design a series noise clipping circuit which rectify the noise signal with amplitude lower than ±VF **(Jan-2020) [NLJIET]**	**7**	**CO1**	**EL**

CHAPTER: 3
BIPOLAR JUNCTION TRANSISTORS

BJT operation, BJT voltages and currents, BJT amplification, BJT switching, CB, CE and CC characteristics, Transistor testing.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	State advantage of transistor **(June-2019) [NLJIET]**	**3**	**CO2**	**RM**
2	Why are junction transistors called bipolar devices? **(Aug-2022) [NLJIET]**	**3**	**CO2**	**UN**
3	Draw the symbol of NPN and PNP transistor. What is use of transistor? **(June-2019) [NLJIET]**	**3**	**CO2**	**CR**
4	Briefly explain regions of operation for BJT. **(Jan-2024) [NLJIET]**	**3**	**CO2**	**UN**
5	Draw the symbol of PNP transistor and NPN transistor and explain why it is called "Bipolar Transistor"? **(Jan-2025-New) [NLJIET]**	**3**	**CO2**	**CR**
6	Discuss about leakage current in transistor. **(Aug-2023) [NLJIET]**	**3**	**CO2**	**CR**
7	Drive the relation between current gain α and β for CE configuration. **(July-2024) [NLJIET]**	**3**	**CO2**	**EL**
8	Derive the relation between current gain α and β **(Jan-2019) (Mar-2023) (Aug-2023) [NLJIET]**	**3**	**CO2**	**EL**
9	Derive relation between α & β for BJT. **(Jan-2024) [NLJIET]**	**3**	**CO2**	**EL**
10	Derive relation between α and β. **(June-2025-New) [NLJIET]**	**3**	**CO2**	**EL**
11	Derive relationship between α dc and β dc of a transistor. **(Jan-2025-New) [NLJIET]**	**3**	**CO2**	**EL**

12	Explain the working of Transistor as Switch **(Jan-2020) [NLJIET]**	**3**	**CO2**	**UN**
13	Draw the symbol of NPN & PNP transistor. Also state the advantage of transistor. **(July-2024) [NLJIET]**	**4**	**CO2**	**CR**
14	Draw the symbol of NPN & PNP transistor, write its different operating regions and also explain why it is called as “Bipolar Transistor” **(Aug-2023) [NLJIET]**	**4**	**CO2**	**CR**
15	The metal lead of the p-side of a p–n diode is soldered to the metal lead of the p-side of another p–n junction diode. Will the structure form an n–p–n transistor? If not, why? **(Aug-2022) [NLJIET]**	**4**	**CO2**	**EL**
16	What is an Early effect, and how can it account for the CB input characteristics? **(Aug-2022) [NLJIET]**	**4**	**CO2**	**UN**
17	The value of alpha increases with the increasing reverse-bias voltage of the collector junction. Why? **(Aug-2022) [NLJIET]**	**4**	**CO2**	**AN**
18	Give comparison between CE, CB and CC configuration of transistor. **(Jan-2019) [NLJIET]**	**4**	**CO2**	**AP**
19	Compare between CB, CE, and CC configurations in BJTs with diagram. **(June-2025-New) [NLJIET]**	**4**	**CO2**	**AN**
20	Compare different BJT configurations. **(Jan-2024) [NLJIET]**	**4**	**CO2**	**AN**
21	Explain transistor as a switch. **(Aug-2023) [NLJIET]**	**4**	**CO2**	**UN**
22	Explain use of transistor as a switch. **(Apr-2022) [NLJIET]**	**4**	**CO2**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain biased and unbiased NPN and PNP transistor in detail. **[NLJIET]**	**7**	**CO2**	**AN**
2	Discuss Common Base (CB) configuration of transistor and explain its input and output characteristics. **(Aug-2023) [NLJIET]**	**7**	**CO2**	**AN**

3	Draw CB configuration and discuss its input and output characteristics with ICBO, ri, ro, and current gain α. **(Jan-2024) [NLJIET]**	**7**	**CO2**	**CR**
4	Draw the circuit of CB configuration of transistor and explain Input and output characteristics of CB configuration of transistor. **(Jan-2025-New) [NLJIET]**	**7**	**CO2**	**CR**
5	Draw and explain CB configuration with input and output Characteristics **(June-2025-New) [NLJIET]**	**7**	**CO2**	**CR**
6	Draw CE configuration and discuss its input and output characteristics with ICEO, ri, ro, and current gain β. **(Jan-2024) [NLJIET]**	**7**	**CO2**	**CR**
7	Draw the circuit of transistor in CE configuration. Sketch the output characteristics and explain active, saturation and cut-off regions. **(June-2019) [NLJIET]**	**7**	**CO2**	**CR**
8	Draw and explain CE configuration with input and output Characteristics. **(June-2025-New) [NLJIET]**	**7**	**CO2**	**CR**
9	Draw and explain input and output characteristic of transistor in common emitter configuration. **(Jan-2019) (Mar-2023) (July-2024) [NLJIET]**	**7**	**CO2**	**CR**
10	Draw CE transistor configuration and give its input and output characteristics. Also derive the relation between current gain of CE, CB and CC configurations.**[NLJIET]**	**7**	**CO2**	**CR**
11	Draw the circuit of CE configuration of transistor. Explain Input and output characteristics. Derive $\alpha = \beta / \beta+1$.**[NLJIET]**	**7**	**CO2**	**CR**
12	Sketch the circuit of the common collector mode of BJT and its output characteristics. Derive the expression for the collector current and gain **(Aug-2022) [NLJIET]**	**7**	**CO2**	**CR**
13	Discuss Common Collector (CC) configuration of transistor and explain its input and output characteristics. **(Aug-2023) [NLJIET]**	**7**	**CO2**	**AN**
14	Compare CB and CC configuration with respect to different transistor characteristics. **(Mar-2023) [NLJIET]**	**7**	**CO2**	**AN**

15	Compare CE, CB and CC configuration with respect to different transistor characteristics. **(June-2019) (July-2024) [NLJIET]**	**7**	**CO2**	**AN**
16	Compare CE, CB and CC configuration with respect to different transistor **(Jan-2025-New) [NLJIET]**	**7**	**CO2**	**AN**
17	Compare CE, CB and CC configuration in details. **(Apr-2022) [NLJIET]**	**7**	**CO2**	**AN**
18	Explain application of transistor as a switch. **(Jan-2019) (Mar-2023) (July-2024) [NLJIET]**	**7**	**CO2**	**UN**

CHAPTER: 4

BJT BIASING

DC load line and bias point, Base bias, Collector-to-base bias, Voltage- divider bias ,Comparison of basic bias circuits, Bias circuit design.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is DC load line? Explain with necessary diagram. **(Jan-2025-New) [NLJIET]**	3	CO2	UN
2	Explain about DC load line of BJT. **(Apr-2022) [NLJIET]**	3	CO2	UN
3	Explain about DC load line of transistor. **(Mar-2023) [NLJIET]**	3	CO2	UN
4	Explain about DC load line and Bias point of transistor **(Jan-2020) [NLJIET]**	3	CO2	UN
5	Why biasing circuits are required? **(Jan-2019) [NLJIET]**	3	CO2	AN
6	Describe the need of biasing the transistor. **(Aug-2023) [NLJIET]**	3	CO2	AN
7	State the importance of transistor biasing. **(June-2025-New) [NLJIET]**	3	CO2	RM
8	What is stability factor? Explain. **(Jan-2019) [NLJIET]**	3	CO2	RM
9	Explain thermal stability on BJT. **(Apr-2022) [NLJIET]**	3	CO2	UN

10	What is the thermal runaway in transistors, and how can it be avoided **(Aug-2022) [NLJIET]**	**3**	**CO2**	**RM**
11	What is DC load line? Explain with necessary diagram. **(Jan-2019) [NLJIET]**	**4**	**CO2**	**UN**
12	Explain the selection of a Q point for a transistor bias circuit and discuss the limitations on the output voltage swing. **(Mar-2023) (Jan-2020) [NLJIET]**	**4**	**CO2**	**UN**
13	Describe the factors affecting the Stability of Q point. **(Aug-2023) [NLJIET]**	**4**	**CO2**	**RM**
14	Write a short note on Thermal runaway. **(Aug-2023) [NLJIET]**	**4**	**CO2**	**RM**
15	Explain fixed bias and self bias in a transistor. **(Jan-2025-New) [NLJIET]**	**4**	**CO2**	**UN**
16	Explain voltage divider biasing of BJT. **(Apr-2022) [NLJIET]**	**4**	**CO2**	**UN**
17	Explain voltage divider biasing with neat diagram. **(June-2025-New) [NLJIET]**	**4**	**CO2**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Discuss load line, Q point, DC & AC load lines for BJT. **(Jan-2024) [NLJIET]**	**7**	**CO2**	**AN**
2	What are the different method for biasing the transistor? Explain any two method with necessary circuit diagram. **(Jan-2019) [NLJIET]**	**7**	**CO2**	**RM**
3	Explain DC load line and Q-point for any transistor configuration. Also state the necessity of biasing and list biasing methods for transistor. **[NLJIET]**	**7**	**CO2**	**UN**
4	Why biasing is important in transistor? Explain voltage divider bias with diagram. **(June-2019) (Mar-2023) [NLJIET]**	**7**	**CO2**	**AN**
5	Draw the fixed-biased circuit by considering an n–p–n transistor in the CE mode. Derive the expressions for stability factors. What are the functions of the coupling capacitors? **(Aug-2022) [NLJIET]**	**7**	**CO2**	**CR**

6	Draw collector to base bias circuit and explain its operation. Also state advantages and disadvantages of the circuit. **[NLJIET]**	**7**	**CO2**	**CR**
7	Draw circuit for Emitter feedback bias and explain in detail with its advantages and disadvantages. **(Jan-2024) [NLJIET]**	**7**	**CO2**	**CR**
8	What is self-bias? Draw the circuit showing self-bias of an n–p–n transistor in the CE mode. Explain physically how the self-bias improves stability. **(Aug-2022) [NLJIET]**	**7**	**CO2**	**RM**

CHAPTER: 5
AC ANALYSIS OF BJT CIRCUITS

Coupling and bypass capacitors, AC load lines, transistor models and parameters, CE circuit analysis, CE circuit with unbypassed emitter resistor, CC circuit analysis, CB circuit analysis, Comparison of CE, CB and CC

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is use of coupling and bypass capacitor? **(June-2019) (Apr-2022) [NLJIET]**	**3**	**CO3**	**RM**
2	Define the use of Coupling capacitor. **(July-2024) [NLJIET]**	**3**	**CO3**	**RM**
3	What is the ac load line in the transistor? Write its significance. **(Aug-2022) [NLJIET]**	**3**	**CO3**	**RM**
4	Explain the AC load line of transistor. **(Jan-2020) (Jan-2025-New) [NLJIET]**	**3**	**CO3**	**UN**
5	Explain the properties and application of common base amplifier. **(July-2024) [NLJIET]**	**3**	**CO3**	**UN**
6	List out the salient feature of emitter follower. **(Jan-2019) (July-2024) [NLJIET]**	**3**	**CO3**	**RM**

7	Briefly explain the effect of coupling capacitor and bypass capacitor on small signal transistor amplifier. **[NLJIET]**	4	CO3	UN
8	Explain AC load line with respect to BJT **(June-2019) (Apr-2022) [NLJIET]**	4	CO3	UN
9	Explain various properties of CB amplifier. **(Jan-2019) [NLJIET]**	4	CO3	UN
10	Explain the working of Emitter follower. **(Aug-2023) [NLJIET]**	4	CO3	RM

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Draw and explain the transistor a.c. equivalent circuit. **(Jan-2019) [NLJIET]**	7	CO3	CR
2	Explain in detail different models used for AC analysis of BJT circuits. **(Aug-2023) [NLJIET]**	7	CO3	UN
3	Determine h-parameters for the two port network. Also draw the hybrid model for CE, CB and CC configurations **[NLJIET]**	7	CO3	AN
4	Draw h parameter equivalent model for CE amplifier and derive the equation for input impedance, output impedance, Voltage gain and current gain for CE amplifier. **(Aug-2023) [NLJIET]**	7	CO3	CR
5	Draw h parameter based equivalent circuit for CE transistor and derive the equation for input impedance, output impedance, voltage gain. **(Apr-2022) [NLJIET]**	7	CO3	CR
6	Derive h parameters for transistor CE amplifier. Also Draw and explain h parameter equivalent model for CE amplifier. **(Jan-2025-New) [NLJIET]**	7	CO3	CR
7	Draw the approximate hybrid model for any transistor configuration at low frequencies. Show that only hie and hfe are essential in the model. Is the approximation justified? **(Aug-2022) [NLJIET]**	7	CO3	CR

8	Briefly explain the h-parameters and draw h-parameter based equivalent circuit for CE transistor and derive equation for input impedance, output impedance and voltage gain. **(Jan-2020) [NLJIET]**	7	CO3	CR

CHAPTER: 6

Field Effect Transistors

Junction Field Effect Transistors, JFET characteristics, JFET data sheets and parameters, FET amplification and switching, MOSFETs

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is FET? State important features of FET. **(June-2019) [NLJIET]**	3	CO4	RM
2	State the advantages of JFETs over BJTs. **(June-2025-New) [NLJIET]**	3	CO4	RM
3	Give comparison of BJT and JFET. **(Jan-2019) [NLJIET]**	3	CO4	AN
4	Compare FET with BJT. **(Jan-2024) [NLJIET]**	3	CO4	AN
5	Describe the operation of a JFET with neat diagram. **(June-2025-New) [NLJIET]**	3	CO4	RM
6	How will you determine the drain characteristics of JFET? What do they indicate? **(Aug-2022) [NLJIET]**	3	CO4	EL
7	Discuss about transconductance curve of JFET. **(Aug-2023) [NLJIET]**	3	CO4	RM
8	Explain trans-conductance and switching in FET. **(Apr-2022) [NLJIET]**	3	CO4	UN
9	Write applications of JFET.**(Aug-2023) [NLJIET]**	3	CO4	RM
10	Explain in brief applications of JFET. **(Jan-2025-New) [NLJIET]**	3	CO4	UN
11	Explain FET as an Amplifier. **(Mar-2023) [NLJIET]**	3	CO4	UN

12	What is MOSFET device? Draw its construction diagram. **(Aug-2022) [NLJIET]**	**3**	**CO4**	**RM**
13	What are the advantage of N-Channel MOSFET over P-Channel MOSFET **(Jan-2019) [NLJIET]**	**3**	**CO4**	**RM**
14	State advantage, disadvantage and application of FET. **(July-2024) [NLJIET]**	**4**	**CO4**	**RM**
15	Compare BJT and FET **(June-2019) (Apr-2022) [NLJIET]**	**4**	**CO4**	**AN**
16	Give Comparison of BJT and FET. **(July-2024) [NLJIET]**	**4**	**CO4**	**AN**
17	Explain (i) Unipolar device (ii) Transconductance **(June-2019) (Mar-2023) [NLJIET]**	**4**	**CO4**	**UN**
18	Explain Transconductance and switching in FET. **(Jan-2020) (Jan-2025-New) [NLJIET]**	**4**	**CO4**	**UN**
19	Justify the name of Depletion type MOSFET. **[NLJIET]**	**4**	**CO4**	**AN**
20	Explain D-type MOSFET with neat diagram. **(Jan-2025-New) [NLJIET]**	**4**	**CO4**	**UN**
21	Explain FET as a switch. **(Jan-2024) [NLJIET]**	**4**	**CO4**	**UN**
22	Explain the common drain configuration for a JFET. **(Aug-2022) [NLJIET]**	**4**	**CO4**	**UN**
23	Explain FET as an amplifier. **(Jan-2024) (Apr-2022) (Jan-2020) (Jan-2025-New) [NLJIET]**	**4**	**CO4**	**UN**
24	Explain the application of FET as a buffer amplifier. **(Jan-2019) [NLJIET]**	**4**	**CO4**	**UN**
25	Draw MOSFET and describe the working. **(June-2025-New) [NLJIET]**	**4**	**CO4**	**RM**
26	Draw symbol of MOSFETS. how enhancement MOSFET is different from depletion MOSFET. **(June-2025-New) [NLJIET]**	**4**	**CO4**	**CR**
27	Write short notes on the following : (i) Advantages of JFET (ii) Difference between MOSFET and JFET **(Aug-2022) [NLJIET]**	**4**	**CO4**	**RM**

28	What are the advantage of N-Channel MOSFET over P-Channel MOSFET **(Mar-2023) [NLJIET]**	**4**	**CO4**	**RM**
29	Write a short note on E MOSFET as an Amplifier. **(Jan-2020) [NLJIET]**	**4**	**CO4**	**RM**
30	Explain E-type MOSFET with neat diagram. **(Jan-2025-New) [NLJIET]**	**4**	**CO4**	**UN**
	DESCRIPTIVE QUESTIONS			
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Compare the different characteristics of the following semiconductor devices: bipolar junction transistor, field-effect transistor. **(Aug-2022) (Jan-2025-New) [NLJIET]**	**7**	**CO4**	**AN**
2	Explain in details the construction and working of n-channel JFET. **(Apr-2022) [NLJIET]**	**7**	**CO4**	**UN**
3	Explain the Depletion region and drain characteristics of n channel JFET. **(Mar-2023) (Jan-2020) [NLJIET]**	**7**	**CO4**	**UN**
4	Explain the construction, working and drain characteristics of N- Channel JFET. **(Aug-2023) [NLJIET]**	**7**	**CO4**	**UN**
5	Draw and explain various characteristic of JFET. **(Jan-2019) [NLJIET]**	**7**	**CO4**	**CR**
6	Draw structure of n-channel JFET and explain its working. **[NLJIET]**	**7**	**CO4**	**CR**
7	Give constructional details of JFET and give its characteristics. Why FET is called voltage controlled device? **[NLJIET]**	**7**	**CO4**	**AN**
8	Write short note on JFET **(June-2019) [NLJIET]**	**7**	**CO4**	**RM**
9	Explain the JFET parameters and establish the relationship between them **(Aug-2022) [NLJIET]**	**7**	**CO4**	**UN**
10	Explain Drain characteristics and Transfer characteristics of JFET in detail with all related terms as Transconductance gm, drain resistance rd, and amplification factor μ. **(Jan-2024) [NLJIET]**	**7**	**CO4**	**UN**
11	Discuss MOSFET in details. **(July-2024) [NLJIET]**	**7**	**CO4**	**AN**

12	Write short note on MOSFET. **(June-2019) (Apr-2024) [NLJIET]**	7	CO4	RM
13	Compare BJT with FET and explain D MOSFET. **(Jan-2020) [NLJIET]**	7	CO4	UN
14	Write a short note: E-Type MOSFET **(Jan-2019) (Mar-2023) [NLJIET]**	7	CO4	RM
15	Describe briefly the construction and working of Enhancement and Depletion mode MOSFET **[NLJIET]**	7	CO4	RM
16	Explain the construction and principle of operation of Enhancement type P-channel MOSFET. **(Aug-2023) [NLJIET]**	7	CO4	UN

CHAPTER: 7

FET biasing

DC load line and bias point, Gate bias, Self bias, Voltage divider bias, Comparison of basic JFET bias circuits

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Why thermal runaway is not there in FETs? **[NLJIET]**	3	CO4	CR
2	Explain the Gate bias circuit used to bias FET. **[NLJIET]**	3	CO4	UN
3	Draw and explain the self-bias circuit of FET. **(Jan-2019) (Mar-2023) [NLJIET]**	4	CO4	CR
4	Draw and Explain DC load line and Bias Point of JFET **[NLJIET]**	4	CO4	CR

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain the voltage divider bias circuit used to bias FET. **[NLJIET]**	7	CO4	UN
2	Explain Voltage divider Biasing of JFET with diagram. **(June-2025-New) [NLJIET]**	7	CO4	UN

3	Explain Self Biasing of JFET with diagram. **(June-2025-New) [NLJIET]**	7	CO4	UN
4	Give a comparison between basic FET biasing circuits. **[NLJIET]**	7	CO4	CR

CHAPTER: 8
SPECIAL PURPOSE DIODES

Light Emitting Diode(LED), Photo diode, Solar cell, PIN diode, Varactor diode, Schottky diode, Tunnel diode, Seven segment display

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain LED diode **(June-2019) [NLJIET]**	3	CO5	UN
2	Advantage, disadvantage and application of LED. **(July-2024) [NLJIET]**	3	CO5	RM
3	Illustrate LED. **(June-2025-New) [NLJIET]**	3	CO5	RM
4	Write principle and applications of light emitting diode. **[NLJIET]**	3	CO5	RM
5	What is varactor diode? How capacitance of a diode varies with reverse voltage? **(June-2019) (Mar-2023) [NLJIET]**	3	CO5	RM
6	Explain the varactor diode. **(Aug-2022) [NLJIET]**	3	CO5	UN
7	Explain Varactor diode and varistor. **(Jan-2020) [NLJIET]**	3	CO5	UN
8	Explain V-I characteristic of tunnel diode. **(Jan-2029) (July-2024) [NLJIET]**	3	CO5	UN
9	Explain Seven segment display. **(June-2025-New) [NLJIET]**	3	CO5	UN
10	Give minimum four comparison of LED and photo diode. **[NLJIET]**	4	CO5	RM
11	Compare PIN diode and Photo Diode. **(July-2024) [NLJIET]**	4	CO5	AN
12	Explain PIN photo diode **(June-2019) (Mar-2023) (Apr-2022) [NLJIET]**	4	CO5	UN

13	Explain the working of PIN Diode. **(Jan-2020) [NLJIET]**	4	CO5	UN
14	Explain the contraction of the solar cell with its operational principle. **(Aug-2022) [NLJIET]**	4	CO5	UN
15	Explain solar cell with diagram. **(Jan-2025-New) [NLJIET]**	4	CO5	UN
16	Explain varactor diode. **(Apr-2022) [NLJIET]**	4	CO5	UN
17	Explain Schottky diode in details. **(July-2024) [NLJIET]**	4	CO5	UN
18	What is a Schottky diode? Explain its advantages over regular diodes. **(June-2025-New) [NLJIET]**	4	CO5	RM
19	Explain the working principle of a Tunnel diode. **(June-2025-New) [NLJIET]**	4	CO5	UN
20	Draw and explain seven segment display. **(Jan-2020) [NLJIET]**	4	CO5	CR
21	Discuss seven segment displays with its types. **(Jan-2024) [NLJIET]**	4	CO5	AN
22	Explain the sixteen segment display and its applications with the necessary circuit diagram. **(Aug-2022) [NLJIET]**	4	CO5	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Describe working principle and applications of PIN Photo diode and Solar cell. **(Jan-2024) [NLJIET]**	7	CO5	AN
2	Discuss the construction and working of a solar cell. Highlight its applications. **(June-2025-New) [NLJIET]**	7	CO5	RM
3	Describe the construction, the symbol, V-I characteristics and application of Tunnel diode. **(Aug-2023) [NLJIET]**	7	CO5	AN
4	Write a short note on Tunnel diode. **(Apr-2022) (Jan-2025-New) [NLJIET]**	7	CO5	RM
5	Draw symbol of tunnel diode. Draw VI characteristic of tunnel diode and explain it. **[NLJIET]**	7	CO5	CR
6	Explain working of Tunnel diode. Draw its I-V characteristic and symbol. List out applications of Tunnel diode. **[NLJIET]**	7	CO5	UN
7	Discuss the construction and applications of a Varactor diode **(June-2025-New) [NLJIET]**	7	CO5	RM

8	Write short note on I. Tunnel diode II. Photodiode III. LED **[NLJIET]**	**7**	**CO5**	**RM**
9	Write a short note on seven segment display with diagram. **(Jan-2025-New) [NLJIET]**	**7**	**CO5**	**RM**

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